



Pankaj Chauhan

SOFTWARE ENGINEER

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“Be the change that you want to see in the world.”

Summary

Software Engineer specializing in Python and Agentic AI, building scalable data extraction pipelines for millions of records. Expert in integrating LLMs, RAG, and custom parser frameworks to achieve 98%+ accuracy while reducing operational and OCR costs.

Experience

Perfios Software Solutions

New Delhi, India

SENIOR MEMBER OF TECHNICAL STAFF

April, 2024 – Present

- **Architected a Corporate Risk platform** to monitor Indian entities and management for adverse media and governance risks. Engineered a 10K/daily pipeline with **custom ML noise-filters** and a scalable API layer. Optimized LLM performance via **advanced prompt engineering**, slashing costs by **85%** and ensuring accurate risk classification (Fraud, Litigation, Misconduct).
- **Engineered a DGFT Trade-Risk pipeline** to monitor export entities via daily court judgment analysis. Combined multi-stage OCR with **prompt engineering** to autonomously classify complex legal risks and extract high-precision compliance parameters.
- Worked on complete data extraction from FIR PDFs spanning all Indian States, covering millions of records.
- Covered diverse formats and achieved **97.5 percent accuracy** in data extraction with **85 percent reduction in OCR cost**.
- Led extensive research on **table extraction** from PDFs and developed a versatile solution for accurate extraction of tables from PDFs and images, resulting in enhanced efficiency and cost savings for more than 10 million records.

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April, 2022 – April, 2024

- Developed an industry-first **PDF Tampering Detection** solution for banking fraud; analyzed 1.5M+ documents and implemented an automated scoring mechanism that slashed false positives, enhanced detection, and restored client confidence.
- **Parser Utility:**
 - Developed a versatile parser utility framework for streamlined extraction of core data fields, tables, and key-value pairs from PDFs, images, and OCR responses, reducing development time by 80-85% while achieving 98.6% data accuracy.
 - Enhanced data accuracy with **OCR Error Management** and **advanced search algorithms**, improving efficiency and reducing latency through **caching** and page cropping to refine regions of interest, resulting in faster processing times.

Karza Technologies Pvt. Ltd.

Mumbai, India

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June, 2021 – April, 2022

- Built low-latency **parser APIs** for KYC services, seamlessly integrating **Google OCR** and **Azure OCR**,
- Achieved a **40% reduction** in API latency by implementing a **caching** mechanism and optimizing searching algorithms.
- Elevated **data accuracy** within APIs by eliminating **OCR errors** and adeptly handling poor orientation in PDFs and images.
- Created custom parsers for complex tabular data, removing the need for expensive Adobe APIs and cutting costs significantly.
- Conducted RCA on API discrepancies and migrated codebases to Python 3.11, ensuring error resolution and compatibility.

SmartServ Inc

Pune, India

SOFTWARE DEVELOPMENT INTERN

Jan, 2021 – June, 2021

- Developed **SQL** and **MongoDB** migration scripts, enhanced front-end with **React JS** and **jQuery**, led bug fixes and managed projects from root cause analysis to production, ensuring quality and timely delivery.

Education

National Institute of Technology Srinagar

Jammu & Kashmir, India

B.Tech. in Information Technology Engineering

July, 2017 – July, 2021

Skills & Expertise

Programming Languages	Python, C++, Java, JavaScript
Generative AI & RAG	Prompt Engineering, Agentic AI, Vector Databases, RAG, Model Context Protocol, LLM Integration
Data Mining	PDF Parsing, Tampering Detection, OCR, Generic Parsers, Pandas, NumPy, PyMupdf, PdfPlumber
Web Technologies	PostgreSQL, MySQL, MongoDB, HTML, CSS, Node.js, Flask
Others	Linux, Git, Jupyter Notebook, VS Code, Jira

Certificates

IBM Specialization	Generative AI for Software Developers [Verify Credential]
IBM Specialization	IBM RAG and Agentic AI [Verify Credential]